

PRECISION SPECTRA AND INITIAL-SET DEPENDENCE IN CONTRACTING CONTROL SYSTEMS

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ABSTRACT. We consider the minimum number of open-loop inputs needed to keep every state of a compact initial set within a Euclidean tolerance of a planar constraint throughout a finite horizon. Two C^2 diffeomorphisms share a contracting origin and make the constraint controlled invariant. Their first derivatives match feasible branch proportions a_0, a_1 with radial factors a_0^β, a_1^β , where $\beta > 1$. For every finite $\gamma \geq 0$ and every vanishing tolerance sequence $\delta_n = \exp(-\gamma n + o(n))$, we determine the full-constraint cost rate as $\max_{p \in [0,1]} H_b(p) \min\{1, \gamma/[\beta \chi(p)]\}$, where H_b is binary entropy and $\chi(p) = -p \log a_0 - (1-p) \log a_1$. Its three explicit pieces have a common-cost boundary $\gamma_* = \beta h_a$, with $h_a = -\sum_i a_i \log a_i$, and a distinct saturation boundary $-(\beta/2) \log(a_0 a_1)$. Every fixed positive-area compact initial set has rate γ/β at and below γ_* . One compact set of arbitrarily nearly full area, chosen before all exponents and sequences, has rate $\min\{h_a, \gamma/\beta\}$ throughout and a strictly smaller rate than the full constraint above γ_* . The geometric foundation is an area bound for every approximately feasible input, including inputs that leave and reenter the constraint. Complete feasible intervals may overlap, have truncated images, or be empty at a fixed radius. Physical bounded-run trajectories with wrong-control margins supply the matching full-set lower bound; stopping scales and type counting evaluate the spectrum.

1. INTRODUCTION

Invariance entropy measures the number of control signals needed to maintain a state constraint; see Kawan [8]. When every input contracts toward a common point, a finite catalogue can suffice for arbitrarily long horizons at each fixed positive tolerance. The catalogue can nevertheless grow exponentially when the tolerance decreases exponentially with the horizon. We study this joint limit and ask whether its cost is shared by all compact initial sets of positive area.

The distinction between typical initial states and the full constraint is essential. A compact set of almost full area can omit rare states whose required controls determine the full-set cost. At the same time, an approximately feasible input need not follow a selected exact code. It may switch between overlapping feasible branches, leave the constraint within the permitted tolerance, and later return. A symbolic count alone does not settle the minimum physical catalogue.

Our main theorem determines the complete full-set spectrum from matched first derivatives at the origin, controlled feasibility of the actual constraint, and a global norm inequality toward the origin. It does not require globally small derivatives, compactly supported perturbations, boundary-ray preservation, an instantaneous relation between branch lengths and radial factors, or full angular images for every feasible word. The resulting spectrum has three pieces. At and below βh_a , where $h_a = -\sum_i a_i \log a_i$, every positive-area compact initial set has rate γ/β . Above it, one fixed compact set of almost full area has rate h_a , whereas the full-set rate increases to $\log 2$. Saturation occurs at $-(\beta/2) \log(a_0 a_1)$, strictly above the common-cost boundary. The near-full set is chosen before all precision exponents, sequences, and horizons.

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Two physical estimates give the result. Theorem 2.3 controls the area covered by any approximately feasible input. Near the origin, a logarithmic slope cone bounds complete feasible intervals without requiring them to map onto the full angular range. At a first true exit, the Euclidean excess gives a band in the actual initial fibre. A finite noncollapsing transient contributes only $O(\delta)$ area. These estimates apply to all feasible prefixes, so no exact strategy is assumed optimal. For the matching full-set lower bound, Lemma 5.1 constructs actual bounded-run trajectories with a uniform margin against the alternative control. Forcing suitably shorter prefixes gives every frequency contribution to the spectrum before its physical margin falls below the tolerance. The upper bound counts all lengths and types in an auxiliary stopping tree, retaining only actual feasible assignments. The exact formula is a consequence of these controlled estimates and classical counting.

Section 7 gives both independently perturbed factors with vanishing overlap and an inward boundary example. In the latter, long constant words have empty feasible slices at each fixed positive radius, although every finite word is realised at sufficiently small radius. A constraint-preserving simultaneous conjugacy cannot reduce this example to the displayed boundary-ray-preserving models. Section 8 treats unequal matching powers in an affine comparison class. Its necessity statement stays within that class. Section 9 gives a smooth collapse example explaining why local matching and all-input convergence alone do not suffice.

1.1. Related work. Our finite-horizon count is the outer invariance complexity of Chen and Zhong [1, Section 2.2], with $n + 1$ constrained times instead of n . Their fixed-tolerance equivalence results concern bounded catalogue size and finite outer equi-invariability. They do not quantify the catalogue constants along a tolerance sequence linked to the horizon. Stabilization entropies [3, 4] study related information costs under prescribed decay or practical stabilization conditions. Here the task is approximate viability of the actual constraint at every time, rather than a prescribed decay estimate for each state.

Measure-theoretic invariance entropy and variational principles are developed in [10, 9]. Chen, Huang and Zhong [2] study entropy dimensions using specified invariant partition cylinders and time-power weights. Their measures need not be invariant. Those cylinder estimates do not themselves bound the physical sets covered by arbitrary approximate inputs at δ_n . The first-exit estimate and the finite transient in Theorem 2.3 supply this additional geometric connection.

Graph transforms and volume distortion are established methods. Kawan and Da Silva [7, Appendix] give volume estimates near partially hyperbolic all-time lifts. In the present class, every bounded bi-infinite trajectory is the origin: iterating the norm inequality from time $-k$ bounds its time-zero norm by θ^k times a fixed bound. The blow-up angular coordinate is singular there and is not a physical unstable bundle. Their fixed-radius volume estimates do not provide the joint tolerance estimate needed here. We use the graph-transform method to prove that estimate directly, including truncated feasible images.

Product weights, stopping scales, types and typical sets are classical tools [6, 5]. Yang, Chen, Yang and Zhou [11] establish cost-threshold pressure and Bowen-equation results for a given invariant partition. Such address counts still require a comparison with all approximate physical inputs before they can determine our catalogue. The contribution here is that comparison through actual area and physical witnesses, rather than a new pressure or type formula. The comparisons with [7, 11] use the published graph-transform appendix and the author versions arXiv:1711.01181v2 and arXiv:2309.01628v3, respectively. The estimates used below are proved in full.

2. THE CATALOGUE AND MAIN RESULTS

Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (2.1) cover K . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first n letters affect this count. Controlled feasibility makes it finite.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. Fix

$$(2.2) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta > 1,$$

and set $r_i = a_i^\beta$, $t_i = a_i^{\beta-1}$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(2.3) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

The main theorem assumes the following properties of the actual maps:

- (1) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 diffeomorphisms, $F_i(0) = 0$ and $DF_i(0) = A_i$.
- (2) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.
- (3) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(2.4) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . No accumulated distortion, word-area bound or catalogue comparison is assumed.

For words write $\lambda_w = \prod_{j=1}^{|w|} a_{w_j}$, with empty product one. Define

$$(2.5) \quad H_b(p) = -p \log p - (1-p) \log(1-p), \quad \chi(p) = -p \log a_0 - (1-p) \log a_1,$$

$$(2.6) \quad S(\gamma) = \max_{p \in [0,1]} H_b(p) \min\{1, \gamma/[\beta \chi(p)]\},$$

where $0 \log 0 = 0$. In particular $H_b(a_0) = \chi(a_0) = h$.

Theorem 2.1 (Precision spectrum and initial-set dependence). *Under the three assumptions above, for every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(2.7) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(2.8) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ/β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(2.9) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 2.2 (Closed spectrum and two thresholds). *Put $\alpha = a_* \in (0, 1/2)$, $c_b = -\log(1 - \alpha)$ and $d_b = \log((1 - \alpha)/\alpha) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(2.10) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.11) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(2.12) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_*$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.

The first threshold concerns equality of rates across all positive-area compact initial sets; the second concerns the full-set catalogue alone. The theorem does not classify high-side rates of arbitrary compact initial sets or assert that only two such rates are possible.

Theorem 2.3 (Area covered by an arbitrary approximate input). *Under the same assumptions, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ one can choose a compact $T \subset K \setminus \{0\}$ of positive area and $C_J < \infty$ so that for every $e \in (0, h)$ there is $C_e < \infty$ such that*

$$(2.13) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_J \delta + C_e \left[e^{-(h-e)m} + \delta e^{(\beta-1)(h+e)m} \right]$$

for every input u , $n \geq J$, $0 \leq m \leq n - J$ and $0 < \delta \leq 1$. The integer and the set precede e, n, δ, u . For $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

3. LOCAL FEASIBLE GEOMETRY

Use (s, y) with $z = sy$, only for $s > 0$. Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 D F_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

We choose the local constants from the maps as follows. Let L bound the Euclidean bilinear operator norm of both second derivatives on $\overline{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*^3$, and shrink $\rho > 0$ so that

$$(3.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i .

In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives, with the generous common constant B ,

$$(3.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example $|N_i| \leq 3$ on this rectangle; quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the displayed choice of B dominates all the constants in (3.3). These are derived local inequalities, not extra hypotheses.

Set $\bar{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \bar{r}^k s_0$. The bounds in (3.2) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(3.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.4). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

3.1. Complete intervals with truncated images. Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(3.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (3.2), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. It is a closed interval, possibly empty or a point. Here is the induction that checks all constraints. The current angular image is an interval, with a graph obeying (3.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (3.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Write $\lambda_w = \prod_{j=1}^{|w|} a_{w_j}$. Along any such feasible graph, (3.3) and the geometric decay of s_j give constants independent of w, s_0 and its initial angle:

$$(3.6) \quad C_R^{-1} s_0 \lambda_w^\beta \leq s_{|w|} \leq C_R s_0 \lambda_w^\beta, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. Integration over the *actual* angular image implies

$$(3.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 7.2 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

3.2. The first true exit. Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least $cs_0\lambda_w^\beta$, by (3.3), (3.6), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(3.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius. Consequently the wrong part of the terminal angular interval has diameter at most $C\delta/(s_0\lambda_w^\beta)$. By (3.6), its initial-angle preimage has length at most

$$C\delta\lambda_w^{1-\beta}/s_0.$$

After integration against $s_0 ds_0$, its physical area is at most

$$(3.9) \quad C\delta\lambda_w^{1-\beta}.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (3.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (3.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (3.9).

4. ALL FEASIBLE PREFIXES AND ACTUAL COVERED AREA

Proof of Theorem 2.3. For any $d > 0$, the total weight $\sum \lambda_w$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$. To see this without importing a new result, regard the weights as Bernoulli probabilities. For a Bernoulli variable X , $\log \mathbb{E}e^{t(X-a_0)} \leq t^2/8$, since the second derivative of this logarithm is a tilted Bernoulli variance, at most $1/4$; apply the exponential Markov inequality with $t = 4d$ and with $-4d$. By (3.7), the area of the union of the corresponding complete feasible regions is summable in m . Overlaps do not invalidate the union bound. Borel–Cantelli for rational $d > 0$ shows that for area-almost every $x \in Q_\rho$, all feasible length- m prefixes at x have zero frequency tending to a_0 , uniformly over those prefixes. Here $Q_\rho = Q \cap \{s \leq \rho\}$. Controlled invariance guarantees at least one such prefix for every m ; the origin can be discarded as an area-zero point.

Choose $J \geq 1$ so large that $\theta^J \sup_Q V \leq \rho$. Every exactly feasible length- J prefix then ends in Q_ρ . For each of the finitely many such words σ , let $D_\sigma \subset Q$ be its complete feasible initial region. Each F_σ is a diffeomorphism, so the pullback of the local exceptional null set is null. Hence on almost every initial state of Q , all exact feasible transient prefixes and all their feasible length- m tails have frequencies tending uniformly to a_0 . The maximum over the finitely many transient choices has this same property. These maxima are measurable: they are maxima over finitely many closed finite-word feasibility regions.

Egorov followed by compact inner approximation gives a compact $T \subset K \setminus \{0\}$ of positive area on which this convergence is uniform. For $K = Q$ it gives $|T| > 1 - \zeta$. This single set is chosen using frequency convergence, without choosing a precision exponent or sequence. For every $\epsilon > 0$ there is $D_\epsilon \geq 1$ such that every feasible tail w , after every feasible σ from every state of T , obeys

$$(4.1) \quad D_\epsilon^{-1}e^{-(h+\epsilon)|w|} \leq \lambda_w \leq D_\epsilon e^{-(h-\epsilon)|w|}.$$

Finitely many short lengths are absorbed in D_e .

The finite transient does not silently assume exact feasibility for an approximate input. The polygon Q has $|Q_\delta \setminus Q| \leq C\delta$ for $0 < \delta \leq 1$. For all prefixes of length at most J , the inverse Jacobian determinants are bounded on the compact $\overline{Q_1}$. If an approximately feasible input first exits during these J steps, its initial state belongs to a pullback of this boundary tube. Summing over its at most J possible first-exit times, with the maximum over the finite prefix family, gives

$$(4.2) \quad |\{x \in Q : \text{this input first exits by time } J \text{ and } x \in A_u(n, \delta; K)\}| \leq C_J \delta.$$

No bounded-distortion assumption is hidden here: the determinant bound exists because these are finitely many actual diffeomorphisms. It is uniform in the continuation, horizon, and input.

Fix an arbitrary input and split its covered states into (4.2) and the states exact through time J . On the latter, let σ be its actual length- J prefix and use F_σ to transport the states to Q_ρ . Its inverse area distortion is a fixed constant. Exact feasibility through the next m steps has area at most $C\lambda_{u_{J+1}\dots u_{J+m}}$, bounded using (4.1). For a first true exit at tail time $j \leq m$, its length- $j-1$ prefix intersects the transported set, so (4.1) applies to that prefix and (3.9) bounds its area. The geometric sum of these bounds is at most $C_e \delta e^{(\beta-1)(h+e)m}$. This proves (2.13), including all times, all inputs, and all later reentries. $\square$

5. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

Lemma 5.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at $(s_L, s_L y_0)$. At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The radius and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. The inverse reference maps are $f_0(y) = a_0 y - a_1$, $f_1(y) = a_1 y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L+1$ digits. Since $f_0(y) + 1 = a_0(y+1)$, $1 - f_1(y) = a_1(1-y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. These radii stay below $\bar{r}^k s_*$, independently of whether the angular recursion has yet been imposed. The scalar map $s \mapsto s R_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (3.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (3.3), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its equation is exactly $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, not a formal symbolic substitution. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (3.3) gives

$$y_k \leq a_0 - a_1 - 3a_0d_L/4 + a_0Bs_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_*d_L/(4a^*)$. Both wrong angular excesses are then at least $d'_L = a_*d_L/(2a^*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system. This proves the lemma, including the physical realisation and the genuine-infeasibility assertion. $\square$

6. MATCHING CATALOGUE BOUNDS AND THE SPECTRUM

The geometric estimates above concern actual states and arbitrary approximate inputs. We now combine them with stopping scales and elementary type counting. The reference tree below is a construction for an upper bound, not a claim about an optimal strategy or a necessary catalogue.

6.1. A full-horizon stopping catalogue. Let $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$ and $c_{\max} = \max_i c_i$. Put $M_0 = 4\sqrt{2}\Lambda C_R\rho$ and

$$t_\delta = \min\{1, (\delta/M_0)^{1/\beta}\}, \quad b_\delta = -\log t_\delta, \quad \Gamma_{n,\delta} = \frac{\beta(b_\delta + c_{\max})}{n}.$$

For $n \geq J$, use the full reference binary tree capped at depth $N = n - J$, stopping at the first $\lambda_w \leq t_\delta$ or at depth N , whichever is earlier.

For each actual initial state choose an exactly feasible infinite input, provided by controlled invariance. Its first J digits give a feasible transient ending in Q_ρ ; the subsequent digits lead to a reference leaf. Keep all transient/leaf pairs with a nonempty actual feasible initial region. Empty regions can be discarded. If a leaf reaches the weight threshold, then (3.6) gives

$$s_{|w|} \leq C_R\rho t_\delta^\beta \leq \delta/(4\sqrt{2}\Lambda).$$

At this exactly feasible state $V = \Lambda s \leq \delta/(4\sqrt{2})$. Append the same infinite zero tail to every such pair. The global inequality (2.4) and $\|x\|_2 \leq \sqrt{2}V(x)$ keep every later Euclidean norm at most $\delta/4$, hence strictly within δ of Q . Before stopping, assigned states are exactly feasible. A horizon leaf already specifies the constraint through time n ; its infinite extension is irrelevant to this horizon. This constructs actual controls for all states, including the origin, and checks every intermediate time even on the appended tail.

For sufficiently small δ , $b_\delta > 0$. The logarithmic cost $C(w) = -\log \lambda_w$ of a threshold leaf lies in $[b_\delta, b_\delta + c_{\max})$, since its proper prefix has cost below b_δ . A horizon leaf not reaching the threshold has cost below b_δ . Thus any nonempty leaf of length $\ell \leq n$, with j zeros and $p = j/\ell$, satisfies

$$(6.1) \quad \ell\chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta\chi(p)]\}.$$

The binomial upper bound $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$ follows by evaluating the corresponding term of the binomial law with parameter j/ℓ . For endpoint types the count is one. Therefore

each leaf type has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including the empty leaf, and at most 2^J transient choices. We obtain

$$(6.2) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

No positive length or full terminal image has been assumed for an actual feasible slice.

Each function under the maximum in (2.6) is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum S is consequently continuous. Eventually $\beta b_{\delta_n} = -\log \delta_n + \log M_0$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient factor and polynomial in (6.2) have zero exponential rate. Hence

$$(6.3) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

We also record the coarser finite-tolerance bound used below. The weights of the reference leaves sum to one by $a_0 + a_1 = 1$, and each is at least $a_* t_\delta$. This identity concerns a reference tree, not a partition of the actual plane. Enlarging a constant to include $n < J$, we have

$$(6.4) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

6.2. Shorter physical prefixes give the lower bound. Suppose first that $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$. Use all length- L blocks starting with zero, ending with one, and containing exactly j_L zeros. Their number and logarithmic weight per digit are

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = \frac{\log M_L}{L}, \quad \chi_L = \chi(j_L/L).$$

All concatenations have runs of length at most L , and every complete block has cost $L\chi_L$. The exact identity

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}$$

and the elementary bounds

$$\frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

show that $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower binomial bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ because the limiting frequency is interior.

Fix L before taking the horizon limit, and choose a strict ratio

$$(6.5) \quad 0 < t < \min\{1, \gamma/(\beta\chi_L)\}, \quad k_n = \lfloor tn/L \rfloor, \quad m_n = Lk_n \leq n.$$

For each of the $M_L^{k_n}$ choices of these initial blocks, append one fixed allowable block forever. Lemma 5.1 realises the resulting sequence as an actual exactly feasible orbit at the same positive radius s_L , independent of n and of the block choices. For every $k \leq m_n$,

$$\lambda_{v|k} \geq \lambda_{v|m_n} = e^{-\chi_L m_n}.$$

If an arbitrary catalogue input first differs from such a witness at time $j \leq m_n$, its preceding actual states coincide with the witness. Their radius at time $j-1$ is at least $C_R^{-1} s_L e^{-\beta\chi_L m_n}$ by (3.6). The wrong radial multiplier is at least $r_*/2$, and Lemma 5.1 supplies an angular excess of at least d'_L . The wrong physical state therefore satisfies

$$(6.6) \quad |z_j| - s_j \geq E_L e^{-\beta\chi_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The strict choice in (6.5) gives

$$\frac{1}{n} \log \frac{E_L e^{-\beta \chi_L m_n}}{2\delta_n} \longrightarrow \gamma - \beta \chi_L t > 0.$$

For large n this contradicts the necessary condition (3.8). A later reentry cannot remove that earlier violation. Thus an approximately feasible input for a witness must agree with all its first m_n digits. Different selected prefixes give distinct actual initial states: at their first different digit one control is exactly feasible and the other has the proved wrong-control margin. A single catalogue input can cover at most one of these witnesses, so

$$(6.7) \quad r_F(n, \delta_n; Q) \geq M_L^{k_n}, \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq tB_L.$$

For fixed L take the supremum over the strict choices of t , then let $L \rightarrow \infty$, and then take the supremum over $p \in (0, 1)$. The endpoint frequencies have zero entropy, so continuity gives

$$\liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma).$$

No uniform positive radius or margin as $L \rightarrow \infty$ is needed: L, s_L, E_L are fixed in each horizon limit. The strict ratio handles critical exponents, integer rounding, and every $o(n)$ fluctuation of $-\log \delta_n$. For $\gamma = 0$, $r_F \geq 1$ and (6.3) give the zero limit. This proves (2.7).

6.3. Positive-area and fixed near-full initial sets. To complete Theorem 2.1, apply Theorem 2.3 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $e \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\beta(h+e)} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-\beta(h+e)m_n}$. Each term on the right of (2.13) is then at most a constant times $e^{-(h-e)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h-e) \min\{1, \gamma/[\beta(h+e)]\}.$$

Letting $e \downarrow 0$ gives the lower bound in (2.8); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (2.7). The evaluation of S below will give the claimed common low-side rate.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 4 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N+1) \exp \left(N \max_{|p-a_0| \leq d_N} H_b(p) \right) = e^{(h+o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. This gives an upper rate h for every precision sequence, without selecting a strategy. Bound (6.4) also gives upper rate γ/β . The lower bound already proved gives (2.9). The construction of T used uniform frequency convergence and compact inner approximation only; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.1 subject to the following elementary evaluation.

6.4. Evaluation of the maximum.

Proof of Corollary 2.2. Reparameterise the maximum by the frequency q of the smaller branch $\alpha = a_*$. This only changes the optimisation variable, not the dynamics or control assignments. Then $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/\alpha) + (1 - q) \log((1 - q)/(1 - \alpha)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = \alpha$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at α .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (\alpha, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $xH_b(q)/(c_b + d_b q)$. This ratio decreases for $q > \alpha$: the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(\alpha) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $\alpha < 1/2$, $c_b + d_b/2 > c_b + d_b \alpha = h$. The three pieces match at the thresholds, proving (2.10)–(2.11) and the low-side assertion of Theorem 2.1. Subtracting (2.9) gives (2.12). Strict increase of H_b on $(\alpha, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

Corollary 6.1 (Matched linear systems). *The pair (A_0, A_1) has all the conclusions of Theorems 2.1 and 2.3.*

Proof. The angular maps are $(y + a_1)/a_0$ and $(y - a_0)/a_1$. Their feasible domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover the fibre. Put $t = \max_i t_i < 1$, choose $\Lambda = 1 + 4t/(1 - t)$ and set $v = t(1 + 1/\Lambda) < 1$. Then $V(A_i x) \leq \max\{\max_i r_i, v\}V(x)$ globally. The remaining hypotheses are immediate. $\square$

7. APPLICATIONS OF THE CRITERION

7.1. Independent factors and vanishing overlap. This subfamily shows how the criterion applies to separately varying branch lengths and radial factors. Fix a smooth function $0 \leq \kappa \leq 1$ equal to one on $\bar{B}(0, 3/2)$ and with support contained in $B(0, 7/4)$. For example, with $E(v) = e^{-1/v}$ for $v > 0$ and $E(v) = 0$ otherwise, take

$$\kappa(x) = \frac{E(169/64 - |x|^2)}{E(169/64 - |x|^2) + E(|x|^2 - 9/4)}.$$

Its denominator is positive and its support is the closed ball of radius $13/8$. Set

$$C_\kappa = \max\{1, \max_{|\alpha| \leq 2} \|\partial^\alpha(s\kappa)\|_\infty\}, \quad C = 1 + C_\kappa, \quad d = -\log a^*,$$

$$\nu_0 = \frac{1}{C} \min \left\{ \frac{a_*}{1000}, \frac{(\beta - 1)d}{4(1 + 2/a_*)} \right\} > 0.$$

For $0 < \nu < \nu_0$, take $\eta, \xi_0, \xi_1 \in C_c^\infty(B(0, 2))$, zero at the origin, with all partial derivatives through order two bounded by ν . Take $0 \leq u < \nu$ and define at the input state

$$P_0 = a_0 + \eta + us\kappa, \quad P_1 = a_1 - \eta + us\kappa, \quad \mathcal{R}_i = a_i^\beta e^{\xi_i}.$$

Consider

$$\begin{aligned} \tilde{F}_0(s, z) &= (s\mathcal{R}_0, (\mathcal{R}_0/P_0)[z + (1 - P_0)s]), \\ \tilde{F}_1(s, z) &= (s\mathcal{R}_1, (\mathcal{R}_1/P_1)[z - (1 - P_1)s]). \end{aligned} \tag{7.1}$$

The transverse coefficient is $1 - P_i$. No identity $\mathcal{R}_i = P_i^\beta$ is imposed.

Corollary 7.1. *Every pair (7.1) satisfies the hypotheses and conclusions of Theorems 2.1 and 2.3. For $u > 0$, both inputs are feasible on a set with nonempty interior, whose fibre width near the origin is of order us^2 .*

Proof. Put $\mu = C\nu$, $p_- = a_* - \mu$, $p_+ = 1 - p_-$, $r = e^\mu(a^*)^\beta$ and $t = e^\mu(a^*)^{\beta-1}/(1 - \mu/a_*)$. All perturbations $P_i - a_i$ and ξ_i , including their first two derivatives, are bounded by μ . They vanish at the origin. Globally $p_- \leq P_i \leq p_+$, $\mathcal{R}_i \leq r$ and $\mathcal{R}_i/P_i \leq t$. Since $-\log(1 - \mu/a_*) \leq 2\mu/a_*$,

$$\log t \leq -(\beta - 1)d + (1 + 2/a_*)\mu < 0, \quad \log r < 0.$$

For $\Lambda = (1 + t)/(1 - t)$, the transverse coefficients on z and s are at most t , giving the global norm factor $\max\{r, 2t/(1 + t)\} < 1$.

For invertibility use $J_i(s, z) = (s, z - \sigma_i s) = (s, v)$, where $\sigma_0 = -1$, $\sigma_1 = 1$. Both shear norms are less than two. The sheared map is $(s\mathcal{R}_i, v\mathcal{R}_i/P_i)$. For $g_i = A_i^{-1}\tilde{F}_i - \text{id}$,

$$J_i g_i = (s\alpha_i, v\omega_i), \quad \alpha_i = e^{\xi_i} - 1, \quad \omega_i = e^{\xi_i} a_i/P_i - 1.$$

It vanishes outside $B(0, 2)$. On its support $|s| \leq 2$, $|v| \leq 2\sqrt{2}$, and

$$\begin{aligned} |\alpha_i| &\leq e^\mu \mu, & \|\nabla \alpha_i\|_2 &\leq e^\mu \sqrt{2}\mu, \\ |\omega_i| &\leq e^\mu \mu(1 + 1/p_-), & \|\nabla \omega_i\|_2 &\leq 2e^\mu \sqrt{2}\mu(1 + 1/p_-). \end{aligned}$$

Differentiating the two components and applying the inverse shear yields

$$\|Dg_i\|_2 \leq 2e^\mu \mu[(1 + 2\sqrt{2}) + (8 + \sqrt{2})(1 + 1/p_-)] < 200\mu/a_* < 1/5.$$

The final coarse estimate follows from $p_- > a_*/2$, $e^\mu < 3$, and $a_* \leq 1/2$. Thus, for every target b , the equation $x = A_i^{-1}b - g_i(x)$ has a unique solution by contraction on the complete plane. Also $D\tilde{F}_i$ is nonsingular. This proves a smooth global inverse. The zero perturbation values give the required first derivatives.

On Q , $\kappa = 1$ and the angular maps are $Y = \sigma_i + (y - \sigma_i)/P_i(s, sy)$. For $f_s(y) = y - 2\eta(s, sy)$ and $\vartheta = a_0 - a_1$, define $\vartheta_\pm$ by $f_s(\vartheta_\pm) = \vartheta \pm 2us$. The derivative of f_s lies in $[1 - 2\nu s, 1 + 2\nu s]$. The roots are inside $(-1, 1)$ because $0 < P_i < 1$ at the endpoints. The complete feasible domains are $[-1, \vartheta_+]$ and $[\vartheta_-, 1]$, and

$$\frac{4us}{1 + 2\nu s} \leq \vartheta_+ - \vartheta_- \leq \frac{4us}{1 - 2\nu s}.$$

The radial factor is less than one, so there is no cap violation. The domains cover the whole fibre, including the vertex. Theorems 2.1 and 2.3 now apply. $\square$

7.2. Inward boundary images and empty fixed-radius slices. Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1 - t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1 - v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1 + s^2}.$$

Consider the single explicit family

$$(7.2) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their first jets are (2.3). For the stated V ,

$$(7.3) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

Thus the norm inequality is checked globally, not inferred from feasible trajectories.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(7.4) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

Their endpoints lie strictly inside $(-1, 1)$, and their overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. This proves controlled invariance and the actual overlap geometry. Theorem 2.1 applies.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(7.5) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. Thus a description in which every word has a nonempty full-image slice does not apply to this example.

This statement concerns $I_0^m(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_0^m$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Missing fixed-radius slices must not be confused with removing a word everywhere in Q .

There is also a constraint-preserving conjugacy obstruction. For (7.2), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(7.6) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

For every pair in (7.1), control zero preserves the whole lower edge as a boundary continuum and control one preserves the whole upper edge. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (7.6) onto the corresponding target intersection. Finite versus continuum is impossible. Thus this example is not reducible to the class (7.1) by a fixed Q -preserving simultaneous control homeomorphism, and in particular not by such a C^1 or bi-Lipschitz map. No assertion about unconstrained coordinate changes or every member of the new class is made.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (7.3) gives a global factor at most $67/90 < 1$. Corollary 2.2 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. It is an evaluation of the spectrum, not a claim that missing fixed-radius words require a different count.

8. AN AFFINE MISMATCH COMPARISON

The matched first derivatives have a definite role in an affine comparison class. Here let $\beta_0, \beta_1 > 1$, $\rho_i = a_i^{\beta_i}$, and define

$$(8.1) \quad \begin{aligned} B_0(s, z) &= (\rho_0 s, (\rho_0/a_0)(z + a_1 s)), \\ B_1(s, z) &= (\rho_1 s, (\rho_1/a_1)(z - a_0 s)). \end{aligned}$$

Set $\chi_R = -\sum_i a_i \log \rho_i$, $\gamma_0 = -\log \max_i \rho_i$, and let $D > 0$ solve $\rho_0^D + \rho_1^D = 1$. The following comparison is restricted to (8.1); it is not a classification of the nonlinear class in Theorem 2.1.

Proposition 8.1. *If $\beta_0 \neq \beta_1$, then for each $\zeta \in (0, 1)$ one can choose a single compact $K_\zeta \subset Q$ of area greater than $1 - \zeta$ such that every precision sequence of exponent $0 < \gamma < \gamma_0$ satisfies*

$$\lim_n \frac{\log r_B(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_B(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R, \quad D > h / \chi_R.$$

Every fixed positive-area compact set has lower rate at least $\gamma h/\chi_R$ in this range.

Proof. Both maps are invertible and contract toward the origin in a norm $\max\{\Lambda|s|, |z|\}$ for large Λ , since $\rho_i/a_i = a_i^{\beta_i-1} < 1$. The complete angular maps are $(y + a_1)/a_0$ and $(y - a_0)/a_1$, with inverse branches $a_0y - a_1$ and $a_1y + a_0$. Every complete prefix-free family partitions each fibre with all endpoints included. For $R_w = \prod_j \rho_{w_j}$, the radial product is exactly R_w and the feasible interval has length $2\lambda_w$, independently of initial radius. Its physical area is λ_w .

We first compare all approximate inputs with radial stopping words. Let $\mathcal{L}_R(n, t)$ stop at the first $R_w \leq t$, or at depth n , and put $N_R(n, t) = \#\mathcal{L}_R(n, t)$. For a first difference from a chosen exact code at time j , common prefix a , the angular separating point b_a satisfies

$$|z_j| - s_j = sR_a(\rho_i/a_i)|y - b_a|/\lambda_a.$$

Thus approximate feasibility implies a one-sided band

$$(8.2) \quad |y - b_a| \leq C_b \delta \lambda_a / (sR_a), \quad C_b = 2 \max_i a_i / \rho_i.$$

At a tie the excess is zero and the band includes it. Integrating against $s ds$ gives area at most $C_b \delta \lambda_a / R_a$. Reentry does not change this necessary condition at time j .

Take midpoint witnesses at radius one for leaves with threshold $t = \delta$, and write $\rho_* = \min_i \rho_i$. Each leaf has $R_w > \rho_* t$. For descendants $w = av$ of a fixed proper prefix a , $\lambda_v \geq R_v = R_w / R_a$, so their intervals have length greater than $2\lambda_a \rho_* t / R_a$. The ordered midpoints below a have at least this spacing. For any input and one first-difference time, (8.2) at radius one therefore contains at most $B = \lceil 1 + C_b / (2\rho_*) \rceil$ such witnesses. There is at most one leaf which is a prefix of the input. Summing over first-difference times proves

$$(8.3) \quad \frac{N_R(n, \delta)}{1 + Bn} \leq r_B(n, \delta; Q) \leq N_R(n, \delta / C_0)$$

for a fixed $C_0 > 1$. For the upper bound, use every leaf as an exact prefix and append the same zero tail. Its complete interval covers the assigned states at all prefix times. An early leaf has radius at most δ / C_0 ; choose C_0 large enough that the origin-contracting norm keeps the entire tail within Euclidean norm $\delta/2$. A depth- n leaf needs no later constraint. This proves the full-time comparison for arbitrary inputs.

When $\gamma < \gamma_0$, every branch crosses both thresholds before the horizon for all sufficiently large n , because $R_w \leq e^{-\gamma_0 n}$ at depth n . In the uncapped tree the weights R_w^D sum to one, and each leaf has $\rho_* t < R_w \leq t$. Hence

$$t^{-D} \leq N_R(t) \leq (\rho_* t)^{-D}.$$

Equation (8.3) gives the full-set rate.

For the smaller initial set, the exact code has cylinder area λ_w . The same Bernoulli type estimate used in Section 4, followed by Egorov and compact inner approximation, gives one compact $T \subset Q \setminus \{0\}$ of area greater than $1 - \zeta$ with uniformly typical frequencies. For every $\epsilon > 0$ there is $A_\epsilon \geq 1$ such that on all its exact prefixes

$$\begin{aligned} A_\epsilon^{-1} e^{-(h+\epsilon)k} &\leq \lambda_w \leq A_\epsilon e^{-(h-\epsilon)k}, \\ A_\epsilon^{-1} e^{-(\chi_R+\epsilon)k} &\leq R_w \leq A_\epsilon e^{-(\chi_R-\epsilon)k}. \end{aligned}$$

The set is chosen before ϵ and all precision sequences. Any positive-area compact initial set contains a positive-area compact set with the same property. Splitting the portion covered by an arbitrary input into agreement through m and first difference at $j \leq m$, and using (8.2), gives

$$\text{area}(T \cap A_u(n, \delta; Q)) \leq C_\epsilon [e^{-(h-\epsilon)m} + \delta e^{(\chi_R-h+2\epsilon)m}].$$

Here $\chi_R > h$. Take $m = \min\{n, \lfloor (-\log \delta) / (\chi_R + \epsilon) \rfloor\}$. Since $\gamma < \gamma_0 \leq \chi_R$, the covering lower rate is $\gamma(h - \epsilon) / (\chi_R + \epsilon)$, and $\epsilon \downarrow 0$ gives $\gamma h / \chi_R$.

Set $K_\zeta = T \cup \{0\}$. For its upper rate, retain all radial stopping prefixes realised on T at threshold δ/C_0 and append the full tails already constructed. With $c_{\max} = -\log \rho_*$, their lengths satisfy

$$\ell \leq \frac{-\log \delta + \log C_0 + c_{\max} + \log A_e}{\chi_R - e}.$$

At each length ℓ their weights are at least $A_e^{-1}e^{-(h+e)\ell}$, whereas all length- ℓ weights sum to one. There are at most $A_e e^{(h+e)\ell}$ such words. Summing up to the displayed length and including an input for the origin gives upper rate $\gamma(h+e)/(\chi_R - e)$. Let $e \downarrow 0$ to obtain the matching limit on this single set.

Finally put $b_i = \rho_i^D$. Since $b_0 + b_1 = 1$,

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

The inequality follows from $-\log t \geq 1 - t$, with equality only if $b_i = a_i$ for both i . Equality would require $\beta_0 D = \beta_1 D = 1$, contradicting mismatch. This proves the strict difference. $\square$

Corollary 8.2. *Within (8.1), all fixed positive-area compact initial sets share a rate on some nonempty interval of precision exponents adjacent to zero if and only if $\beta_0 = \beta_1$.*

Proof. Under equality use Corollary 6.1 and Theorem 2.1. Under inequality use Proposition 8.1 for every $0 < \gamma < \gamma_0$. $\square$

9. WHY LOCAL MATCHING ALONE IS INSUFFICIENT

The following example keeps the common attracting origin and even agrees with the linear maps on a neighbourhood of it. It violates global injectivity by collapsing positive-area regions onto an invariant ray.

Proposition 9.1. *For every choice of a_0, a_1, β in (2.2) and every $0 < \sigma < 1/2$, there are C^∞ maps $S_0, S_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with these properties:*

- (1) *They equal A_0, A_1 on $\{s \leq \sigma\}$, so all derivatives at the origin agree.*
- (2) *All inputs drive bounded sets uniformly exponentially to the origin, and Q is controlled invariant.*
- (3) *The compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and $r_S(n, \delta; K_\sigma) = 1$ for every n and every $\delta > 0$.*
- (4) *For every finite $\gamma \geq 0$ and every sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$, the full-set cost satisfies*

$$\min\{h, \gamma/\beta\} \leq \liminf \frac{\log r_S(n, \delta_n; Q)}{n} \leq \limsup \frac{\log r_S(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 < \gamma \leq \beta h$ the full-set rate is γ/β , but the positive-area set K_σ has rate zero.

Proof. Let $e(t) = 0$ for $t \leq 0$ and $e(t) = \exp(-1/t)$ for $t > 0$, and define

$$\psi(s) = \frac{e(s - \sigma)}{e(s - \sigma) + e(2\sigma - s)}.$$

The denominator is positive everywhere. Thus ψ is smooth, zero for $s \leq \sigma$, and one for $s \geq 2\sigma$. Put $\rho_i = a_i^\beta$ and define

$$(9.1) \quad S_0(s, z) = (\rho_0 s, (1 - \psi(s))a_0^{\beta-1}(z + a_1 s) - \psi(s)\rho_0 s),$$

$$(9.2) \quad S_1(s, z) = (\rho_1 s, (1 - \psi(s))a_1^{\beta-1}(z - a_0 s) + \psi(s)\rho_1 s).$$

They have the claimed local identity. Each $S_i(s, z)$ is a convex combination of $A_i(s, z)$ and $R_i(s, z)$, where $R_0 = (\rho_0 s, -\rho_0 s)$ and $R_1 = (\rho_1 s, \rho_1 s)$. All four linear maps contract toward

zero in one adapted norm $\max\{M|s|, |z|\}$ for sufficiently large M : the second-coordinate factor for A_i is at most $a_i^{\beta-1} + a_i^{\beta-1}a_{1-i}/M < 1$, and R_i has factor at most ρ_i . Convexity of the norm gives the same contraction bound for S_i on all of $\mathbb{R}^2$.

For controlled feasibility let $t = \psi(s)$, $y = z/s$, and $\xi = a_0 - a_1$. If $t < 1$ the full feasible intervals are

$$J_0(s) = \left[-1, \min \left\{ 1, \xi + \frac{2a_0t}{1-t} \right\} \right],$$

$$J_1(s) = \left[\max \left\{ -1, \xi - \frac{2a_1t}{1-t} \right\}, 1 \right].$$

They follow by imposing $-1 \leq z'/s' \leq 1$ in (9.1)–(9.2), and their union is $[-1, 1]$. For $t = 1$ both branches are feasible on the whole fibre. The radius always decreases, so Q is controlled invariant.

On K_σ , one application of S_0 sends every state to the lower ray. At every radius $S_0(s, -s) = (\rho_0s, -\rho_0s)$, so input 0^∞ keeps the whole set exactly in Q . Its area is $\int_{2\sigma}^1 2s \, ds = 1 - 4\sigma^2$.

To obtain the full-set lower bound, use $Q_c = Q \cap \{s \leq c\}$ with $c = \sigma/2$. For every input, the radius of a state in Q_c stays at most c . Thus $\psi = 0$ forever along that trajectory, even when its angle leaves $[-1, 1]$, and its entire trajectory is exactly that of A_i . Consequently $r_S(n, \delta; Q) \geq r_A(n, \delta; Q_c)$. Corollary 6.1 and (2.8) give the lower bound.

For the upper bound choose an integer t_0 such that $(\max_i \rho_i)^{t_0} \leq \sigma$. Every initial state admits an exact feasible prefix of length t_0 , ending in the region $s \leq \sigma$. All subsequent trajectories there coincide with the linear system, for all inputs, since the radius continues to decrease independently of the angle. Concatenate all 2^{t_0} possible prefixes with a catalogue for the linear full set over the remaining horizon. For $n \geq t_0$ this gives

$$r_S(n, \delta; Q) \leq 2^{t_0} r_A(n - t_0, \delta; Q) \leq 2^{t_0} C_U \delta^{-1/\beta}.$$

The 2^n exact-prefix bound also holds. These yield the upper rates for every stated precision sequence. Finally $\det DS_i = (1 - \psi(s))a_i^{2\beta-1}$ vanishes on the saturated region, identifying the noninjective mechanism. $\square$

For fixed σ this proposition gives one large zero-cost initial set in one fixed system. Making its area arbitrarily close to one changes σ and the system. This differs from Theorem 2.1, where all K_ζ are chosen inside a single fixed system. The collapse example is a counterexample to sufficiency of local matching and uniform convergence; it is not a counterexample to the diffeomorphism theorem.

10. CONSEQUENCES AND SCOPE

Theorem 2.3 is a bound for actual initial states under every approximately feasible input. It handles overlap through typicality of all feasible prefixes, and handles reentry by retaining the constraint at the first true exit. The finite transient is controlled through inverse Jacobians, rather than an assumption that approximate inputs remain exactly feasible near the origin. No lower bound on every complete word interval is needed.

Theorem 2.1 separates two uses of initial geometry. Every positive-area compact set contains enough typical mass to force the low-side rate. The full-set lower bound instead uses rare physical states at a fixed small positive radius; their wrong-control margins exclude any alternate approximate input. Type counting of all stopping lengths and shorter forced physical prefixes gives the exact spectrum (2.11), including its high-side limit. The auxiliary witness radius may depend on the fixed block length, whereas the near-full compact initial set is fixed before all exponents. This distinction is responsible for the shared-rate boundary.

The criterion retains a particular matched pair of first derivatives, a common attracting origin, controlled feasibility of the triangle, global invertibility and a global norm inequality. It applies beyond the boundary-ray-preserving examples, as the inward system shows. It does not address constant overlap in the limiting angular maps, noninvertible collapse, or arbitrary control systems. The affine mismatch criterion does not establish necessity in the full nonlinear class. The high-side spectrum of arbitrary compact initial sets is not classified, and no fixed-strategy optimality is asserted. In this matched class, nonlinear geometry changes the constants in the controlled estimates, while the exact full-set rate depends only on the displayed first derivatives.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

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